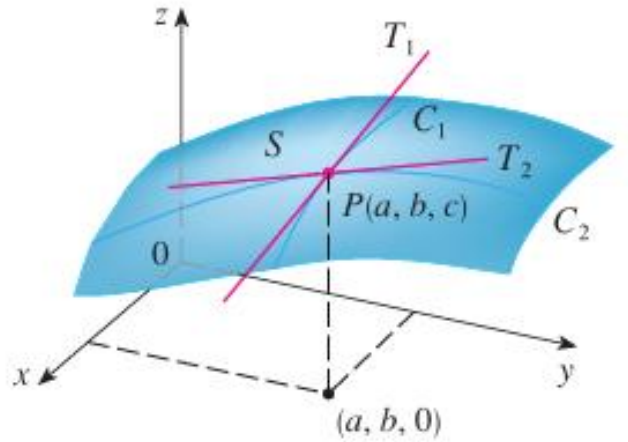
3.3 Partial Derivatives

Motivation: Suppose we have a surface given by the two-variable function $z = f(x, y)$ and a point P (with coordinates of (a, b, c)) on the surface determined by f.



Suppose we want to know the rate at which the z-coordinate changes at P when we are moving across the surface parallel to the x-axis. This means that we would be moving along the curve C_1 toward P in the picture above. At this point P we can imagine a tangent line T_1 being constructed.

Suppose we want to know the rate at which the z-coordinate changes at P when we are moving across the surface parallel to the y-axis. This means that we would be moving along the curve C_2 toward P in the picture above. At this point P we can imagine a tangent line T_2 being constructed.

In Calculus I we found slopes of tangent lines to a curve $y = f(x)$ by calculating the derivative $f'(x)$ and evaluating it at the appropriate point.
In Calculus III for a function $z = f(x, y)$ we also need a way to find a derivative, but in a way that allows us to account for the direction that we are moving along the surface. This leads to the idea of what we call <u>*partial derivatives*</u>.

Note: The information that we are seeking above can be key when one wishes to better understand how quickly a surface is changing at a point.

Definition: If f is a function of two variables, its <u>partial derivatives</u> are the functions f_x and f_y defined by: $f_x(x, y) = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$ and $f_y(x, y) = \lim_{h \to 0} \frac{f(x, y+h) - f(x, y)}{h}$.

Note: $f_x(x, y)$ represents the rate of change of z with respect to x when y is fixed, and $f_y(x, y)$ is the rate of change of z with respect to y when x is fixed.

Note: In the context of the motivation, we would use f_x to find the slope of the line tangent to the brown curve at P, and we would use f_y to find the slope of the line tangent to the green curve at P.

Notation: There are many notations for partial derivatives. We go over 7 possibilities. If $z = f(x, y)$, we write

$$f_x(x,y) = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x,y) = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f$$

$$f_y(x,y) = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x,y) = \frac{\partial z}{\partial y} = f_2 = D_2 f = D_y f$$

Example 1: If $f(x,y) = 2xy^3 + 6xy - x^7$, find $f_x(x,y)$ and $f_y(x,y)$.

Example 2: Consider the surface given by $f(x,y) = 4 - x^2 - 2y^2$. Let C be the curve of intersection between this surface and the plane $x = 2$. Find the slope of the line tangent to C at the point $(2,1,-2)$ on the surface.

Example 3: Find $\frac{\partial z}{\partial x}$ if z is defined implicitly as a function of x and y by $x^3 + y^3 + z^3 + 6xyz = 1$.

Note: There is a fairly subtle point that could be made about the above example. In particular, we assumed the equation defines z implicitly as a function of x and y. What guarantees that the above equation defines z implicitly as a function of x and y? In order to fully answer this question one needs the implicit function theorem (the proof of which is beyond the scope of this course).

Definition: If f is a function of two variables, then f_x and f_y are functions of two variables. The functions $(f_x)_x$, $(f_x)_y$, $(f_y)_x$, and $(f_y)_y$ are called the <u>second partial derivatives of f.</u> If $z = f(x,y)$, we use the following notation:

$$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

$$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 z}{\partial y \partial x}$$

$$(f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 z}{\partial x \partial y}$$

$$(f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

<u>Example 4:</u> Find all four second partial derivatives of $f(x,y) = x^2y^2 - y\sin(x^2)$.

Note: Interestingly, in the above example f_{xy} and f_{yx} are the same! The next theorem addresses this.

Note: As one would probably expect, the second partials f_{xx} and f_{yy} tell us something about the concavity of the curve of intersection between the surface given by $f(x,y)$ and a plane of the form $y = a$ or $x = b$. We will see some more applications of these second partials later in chapter 4.

Note: It is more difficult to perceive what the second partials f_{xy} and f_{yx} tell us. What f_{xy} does communicate is how the slope of the tangent line with respect to x changes as the value of y varies (similarly for f_{yx}). To see an animation of what we mean by this one can visit the following webiste:

https://web.archive.org/web/20180215172659/http://www.math.harvard.edu/archive/21a_fall_08/exhibits/fxy/index.html

Theorem: Suppose f is defined on a disk D that contains the point (a,b). If the functions f_{xy} and f_{yx} are both continuous on D, then $f_{xy}(a,b) = f_{yx}(a,b)$.

Proof: Beyond the scope of this course

Note: In the most recent example, f_{xy} *and* f_{yx} *are continuous on all of* $\mathbb{R}^2$*, and we saw that they were indeed equal.*

We can have partial derivatives of order 3 and higher.

Example 5: If $f(x,y,z) = \sin(3x + yz)$, then find f_{xxy}.